

Anchor

A Windows attention assistant for prevention, recovery, and task continuity

Anchor is built around one practical insight from interviews: distraction is damaging not only because attention moves, but because the original context disappears. The project therefore preserves context before it adds friction, and it gives the user a manual recovery path when software cannot observe an internal lapse.

Decision	Approved direction
Platform	Windows 11 x64
Product form	Background assistant with Goal Beacon, overlays, dashboard, and optional browser adapter
Primary users	People with ADHD and related attention or executive-function difficulties
Core promise	Help the user return to a declared task without claiming to diagnose attention
Build status	Verified hackathon MVP with deterministic replay, tests, and self-contained package

Active prevention + passive recovery + progress tracking + focus toolkit

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1 Interview evidence and product response

The interview notes identify visual over-attention, fragile memory after interruption, sensitivity to sound and artificial light, a need for movement, and the usefulness of a refreshed task cue. These observations point away from a punitive blocker and toward an adaptive assistant that can simplify a scene, preserve working context, and tolerate fidgeting.

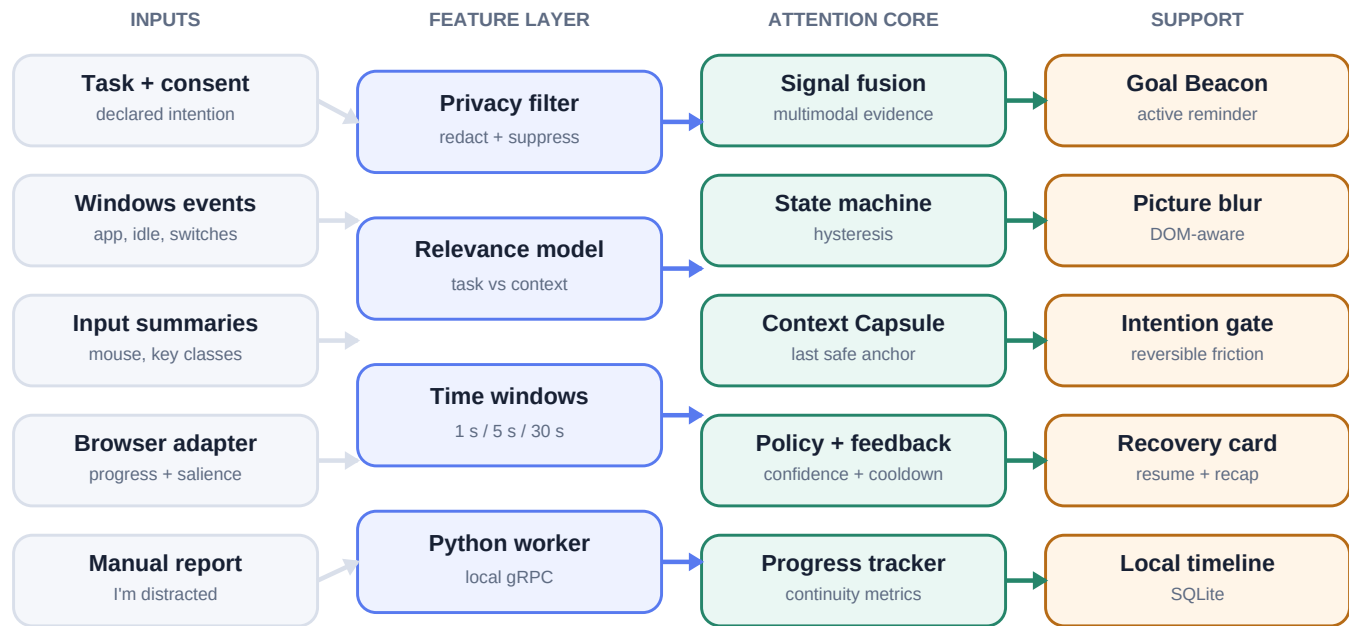
Observed issue	Interpretation	Implemented response
Detailed pictures or animation capture attention	Saliency can overpower task relevance	Dynamic browser image blur, desaturation, visual filter, motion-aware scoring
A distraction causes forgotten context	Working memory is fragile across task switches	Continuous Context Capsule plus automatic and manual Recovery Card
Attention jumps ahead in text	Reading position is not equal to semantic progress	Progress ratio, forward-skip event, future-text mask, recovery anchor
Attention stalls on one phrase	Repeated dwell may signal comprehension friction	Stuck state, phrase event, recap and break-down actions
Random sound or light disrupts work	Environmental stimulation competes with the task	Visual reduction now; audio-transient and sensory presets as scoped extensions
Movement or fidgeting is sometimes helpful	Motion alone is not evidence of failure	Multisignal inference; pointer movement has low weight without corroboration
The user notices distraction first	Internal lapses are not always externally visible	I'm distracted button and Ctrl+Shift+F12 threshold bypass

Product decision

Anchor is a general task assistant with specialized adapters. Reading is the richest current adapter, not the whole product. The user declares a concise intention; Anchor infers supporting context and exposes corrections rather than pretending to know the user's mind.

2 Complete system

The Windows host owns consent, native integration, overlays, safety, and persistence. A local Python worker may improve scoring but cannot restrict input or draw overlays. The browser adapter supplies DOM-level context that a desktop-wide process cannot safely infer.



Safety watchdog releases every restrictive path; all retained data stays local.

Figure 1. Every implemented function participates in the same local feedback loop.



Dismissals raise the intervention threshold; successful returns reduce it slightly.

Figure 2. Closed-loop assistance: preserve context before intervention and learn only from explicit response.

3 Attention detection

Technical complexity comes from combining weak, noisy signals over time rather than relying on a webcam or a simplistic inactivity timer. Raw input is immediately reduced to safe features, then fused through deterministic scoring and temporal hysteresis. The optional Python worker adds calibrated logistic scoring and exponential smoothing over authenticated loopback gRPC.

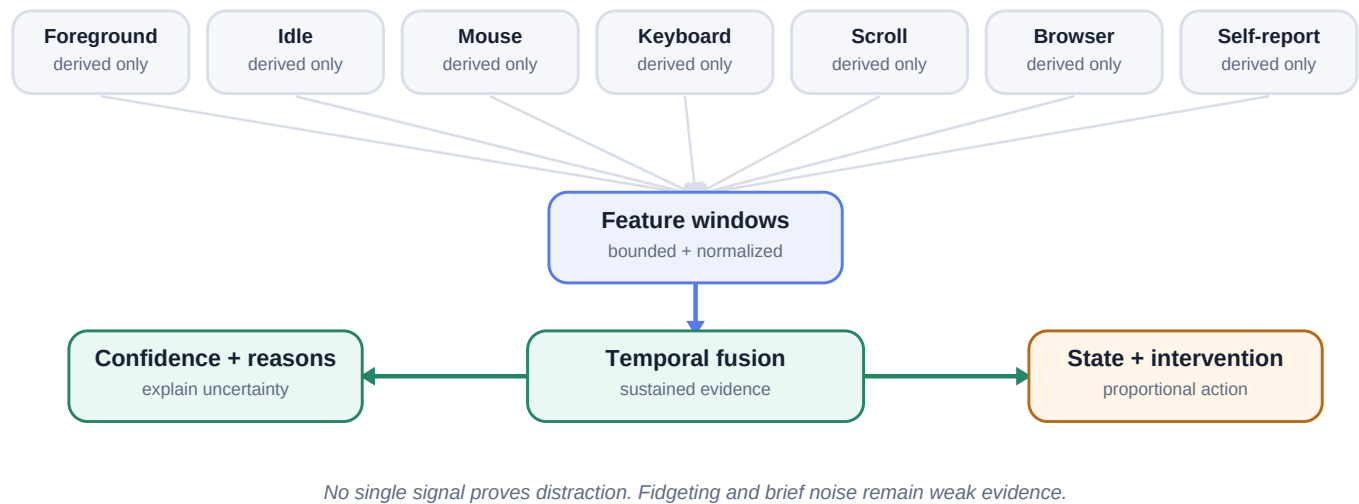
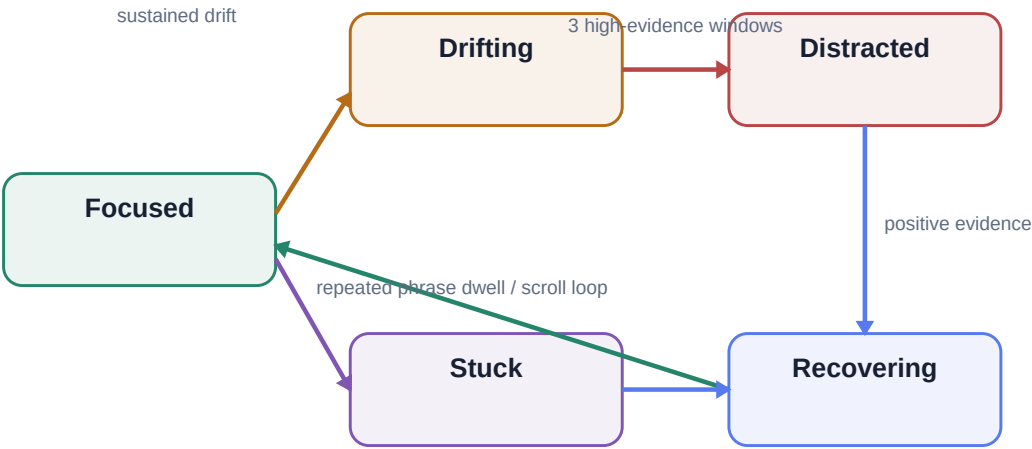


Figure 3. Multisignal attention pipeline. Sensor storms are bounded; stale events may be dropped, but manual reports are preserved.

Mechanism	Derived features	Why it helps	Privacy rule
Foreground desktop	Process, redacted title, app switches, relevance	Distinguishes task-supporting context from probable detours	Secure and system-critical windows suppress intervention
Mouse	Distance and wandering evidence	Finds unproductive roaming only when other signals agree	No coordinates retained
Keyboard	Counts by key category	Typing continuity can support focused evidence	Raw keys and text are discarded
Scroll	Reversal count and repeated loops	Detects skimming loops and stuck reading	No page content required
Idle	Seconds since last input	Supports interruption detection and return timing	Never used alone as proof
Browser	Origin, title, progress, image salience, phrase dwell	Adds semantic reading and visual signals	No browsing-history permission
Manual report	Explicit event	Covers invisible internal distraction	Immediate recovery; no threshold
Optional vision/audio	Head pose, coarse gaze, transient features	Potential corroborating evidence	Opt-in, local, absent from required MVP

4 State model and intervention policy

Anchor exposes uncertainty as an explicit state, confidence value, and reason codes. It takes stronger and sustained evidence to become Distracted than to notice initial drift. Recovery also requires repeated positive evidence, preventing rapid flicker between states.



Manual report jumps directly to Distracted and Recovery Card. Secure context moves to Unknown and clears overlays.

Figure 4. Core state transitions. Unknown is reserved for secure or unavailable classification contexts.

Evidence state	Default response	User control
Focused	Silent observation and Context Capsule refresh	Stop session or change task
Drifting	Short Goal Beacon pulse	Ignore; reduced sensitivity follows repeated dismissals
Distracted	Visual filter and Intention Gate	Return, Continue, Park, Disable, or Esc
Recovering	Recovery Card with last anchor and next action	Resume, Reopen, Recap, Break down, Dismiss
Stuck	Phrase-oriented recovery and task decomposition	Reveal text, request explanation, or continue
Unknown / secure	Clear every active intervention	No content capture

5 Prevention and recovery functions

Function	What it does	How it works	System role
Goal Beacon	Keeps the declared task visible and pulses after sustained drift	Always-on-top WinUI overlay; capped opacity animation	Lowest-friction active prevention
Dynamic Picture Blur	Softens visually dominant page images	Area, animation, contrast proxy, position, and relevance score drive CSS blur	DOM-aware salience reduction
Future-text mask	Prevents attention from jumping far ahead	Masks paragraphs after current progress plus configurable lookahead; hover reveals	Reading prevention
Visual filter	Dims distracting desktop regions	Click-through, topmost overlay that clears on safety signals	Cross-application prevention
Intention Gate	Adds a brief decision before a likely detour	Confidence threshold, cooldown, task prompt, Return/Continue/Park/Disable	Strong but reversible prevention
Pointer guard	Limits pointer movement during a high-confidence gate	Opt-in ClipCursor wrapper with focus-loss and watchdog release	Strongest optional friction
Context Capsule	Preserves where the user was and what comes next	Stores last confident sanitized application, document, location, action, and next step	Passive recovery foundation
Manual recovery	Restores context even when detection misses	Button and global hotkey produce manual_report and bypass thresholds	User-controlled recovery
Recovery Card	Shows anchor, recap, and smallest next step	Deterministic local summary; no cloud model required	Return after interruption or stuck state
Progress timeline	Shows continuity, interruptions, and recovery time	Derived events through ProgressTracker and SQLite WAL store	Feedback without streak pressure

Focus toolkit

The dashboard exposes Dynamic image blur, Hide future text, Visual focus filter, Pointer guard, and Audio shield controls. Browser blur and future-text masking are operational. Audio shield is a user-facing extension point for stable masking sound; microphone capture is not required by the MVP.

6 Technical architecture and libraries

Layer	Libraries and APIs	Responsibilities	Failure behavior
Windows host	.NET 10, C# 14, WinUI 3, Windows App SDK 2.5	Dashboard, overlays, session lifecycle, hotkeys, native message hook	UI stays responsive; restrictions fail open
Core domain	Immutable C# records, Channels, deterministic services	Relevance, temporal evidence, state, policy, recovery, replay	No worker or UI dependency
Windows sensing	SetWinEventHook, Raw Input, GetLastInputInfo, ClipCursor	Foreground, app switches, aggregate input, idle, pointer guard	Secure-window and watchdog release
Persistence	Microsoft.Data.Sqlite, WAL, parameterized commands	Local redacted events and deletion	Bounded payloads and transactions
Worker IPC	gRPC.Net.Client, grpcio 1.84, Protocol Buffers	Random token, protocol health, deadlines, shutdown	Two restarts, then deterministic fallback
Worker inference	Python 3.11-3.12, NumPy	Normalization, logistic scoring, exponential smoothing	Capability flags; optional modules fail independently
Optional vision	OpenCV 4.12+, MediaPipe 0.10+	Future head pose, coarse gaze, saliency	Not installed in core path
Browser	Manifest V3 JavaScript, CSS, activeTab, scripting	Image score, mask, skip/stuck events	Protected pages fail closed
Testing	xUnit, pytest, Node test runner	Unit, integration, protocol, DOM, safety, replay	Deterministic scenarios run offline

Deployment

The build publishes an unpackaged, self-contained x64 executable. Windows App SDK registration-free initialization is enabled explicitly; the single-file executable extracts native runtime dependencies on first launch. Development launch uses the WinApp debug runner to provide temporary identity.

Primary interface contracts

Contract	Purpose
IIInferenceEngine	Starts/stops the worker and returns bounded AttentionPrediction values
ISensorCoordinator	Controls local event collection for one session
IIInterventionPresenter	Shows and clears proportional user support
IEventStore	Appends, queries, and deletes local DerivedEvent records
ReplayScenarioRunner	Feeds JSONL scenarios through the same SessionOrchestrator

7 Privacy, safety, and degraded modes

The project handles attention data as sensitive. Local processing is the default, raw inputs are minimized, and restrictive behavior is never allowed to survive uncertainty or a crash.

Risk	Control	Verification
Raw text or secrets retained	Store key categories and counts only; redact selected text and feature maps	Redaction and serialization tests
Password/payment interference	Secure-window classifier and browser protected-page exclusion	Policy, orchestrator, and DOM tests
Worker hangs or lies	Loopback token, protocol version, deadlines, bounded restart, deterministic fallback	Real-worker and missing-worker tests
Overlay traps the user	Esc, focus loss, secure context, shutdown, process exit, 30-second watchdog	SafetyWatchdog tests and published smoke test
False positives escalate	Sustained evidence, confidence threshold, cooldown, dismissal sensitivity	State and policy tests
History persists unexpectedly	Local SQLite only and Delete local history action	Round-trip and deletion tests
Camera or audio unavailable	Optional capabilities fail independently	Core session requires neither

Degraded-mode ladder

Full local worker + browser adapter -> local worker without browser -> deterministic Windows host -> manual recovery only. Each lower mode remains usable and is labeled in the interface. Anchor never blocks the session because a camera, extension, Python package, or network connection is absent.

Ethical boundary

Anchor is not a medical device, surveillance tool, exam proctor, or productivity score. It should not be used to evaluate another person without their meaningful consent. The interface says why an intervention appeared and offers an immediate way to dismiss or disable it.

8 Hackathon plan and evidence

Workstream	Deliverable	Status
Domain and attention engine	Validated models, relevance, state machine, policy	Complete
Recovery and persistence	Context Capsules, redaction, progress, SQLite	Complete
Worker protocol	Authenticated gRPC, Python scoring, fallback	Complete
Windows integration	Foreground, idle, raw input summaries, watchdog	Complete
Desktop experience	Dashboard, Goal Beacon, filters, gate, recovery	Complete
Browser adapter	Image blur, future text, skip/stuck detection	Complete
Demo and packaging	Three replays, scripts, docs, single-file package	Complete
Stretch research	Gaze/head pose, audio transients, OCR/UIA saliency	Future

Deterministic demo evidence

Scenario	Expected state path	Expected support
Focused to distracted	Focused -> Drifting -> Drifting -> Distracted	BeaconPulse -> BeaconPulse -> IntentionGate
Interrupted and returned	Focused -> Distracted -> Recovering -> Focused	Manual RecoveryCard -> recovery -> quiet
Stuck reading	Focused -> Drifting -> Stuck	RecoveryCard with saved phrase context

Evaluation metrics

For judging, report detector state accuracy on replay, false-intervention dismissal rate, median return time after interruption, percentage of sessions usable without the worker, recovery-card usefulness ratings, and safety-release latency. Avoid presenting an 'attention score' as a clinical fact.

Current automated evidence

The final audit passes 56 .NET tests, 9 Python tests, 6 browser tests, all three end-to-end replay scenarios, Release publish, and a direct launch smoke test of the self-contained executable.

9 Innovation and feasibility assessment

Idea	Innovation	Technical complexity	Hackathon feasibility	Assessment
Gaze-only line reader	Medium	High	Low-medium	Too brittle as the core; keep as optional evidence
Task Beacon	Medium	Low-medium	High	Strong visible demo when tied to sustained drift
Dynamic Picture Blur	High in task-aware form	Medium-high	High in browser	Best prevention differentiator
Intention Gate	Medium-high	Medium	High	More humane than blocking; must remain reversible
Context Capsule	High	High	High with deterministic anchors	Best recovery differentiator
Manual distraction recovery	Medium	Low	Very high	Essential because internal lapses are unobservable
Pointer confinement	Low-medium	Medium-high safety burden	Medium	Opt-in only; demonstrate fail-open controls
Audio therapy / transient detection	Medium	High	Low for MVP	Research extension; avoid unproven clinical claims
Multimodal fusion and replay	High	High	High with deterministic baseline	Core technical-depth story

Recommendation

Lead the pitch with the interruption-recovery loop, not with eye tracking. Show that Anchor preserves the user's place, gently prevents a detour, and still works when automated detection fails. Dynamic Picture Blur supplies a concrete visual moment; Context Capsules and deterministic replay supply the deeper technical and human-centered story.

References

Approved internal design: docs/superpowers/specs/2026-09-19-anchor-general-task-assistant-design.md

Microsoft Windows App SDK self-contained deployment:

learn.microsoft.com/windows/apps/package-and-deploy/self-contained-deploy/deploy-self-contained-apps

Project privacy and demo guides: docs/privacy.md and docs/demo-script.md